

Rules

Rules for cropping, captioning, and layer classification of regions from technical documentation (CAD drawings + text manuals). Loaded verbatim into the propose-agent and re-crop agent system prompts via the rules loader module.

A — Macro-Object Discovery (Pass 1)

Your sole task is to identify and extract COMPLETE, high-level structural units from the page canvas. Do not crop individual small annotations or single balloon callouts; group them as part of their parent macro-regions.

- **A.1. Identify and extract full mechanical views.** Find and crop the boundaries of entire, contiguous drawings. Keep all local component callouts inside the view boundaries.
- **A.2. Extract full specs tables.** Crop entire Bill of Materials (BOM) tables, parameter tables, specification sheets, or legend tables as complete, undivided units. Slicing through a table is strictly forbidden. **CO-LOCATION & OVERLAP EXCEPTION (MANDATORY):** When a BOM table, specs sheet, technical notes block, and title block are stacked directly on top of each other or share adjacent boundaries, you are **strongly encouraged to generously overlap their boundaries** to ensure no text or lines are ever sliced. If they cannot be cleanly separated, **group and crop the entire stacked specifications column as a single, unified Macro-Object region** (e.g. label 'title_block_bom_table'). This guarantees that no table lines or texts are ever sliced or clipped. **EXCLUSION:** The Table of Contents (TOC) is NOT a specs table; it carries zero machine-catalog value and must be completely ignored and excluded from cropping.
- **A.3. Extract standalone notes blocks.** Group any dense, independent text notes or technical descriptions into a single macro-region.
- **A.4. Commit criteria — positive selection.** Commit a region strictly when it contains a cohesive, complete visual structure (a table, a drawing, or a complete specs block).
- **A.5. Skip criteria.** Ignore vast areas of completely empty white space, decorative borders, page headers, or pages containing only running paragraphs of prose narrative with no visual structures.
- **A.6. Exclude part balloons on macro borders.** Individual part balloon callouts (circled numbers pointing to components) or leader lines that reside on the outer margins of a mechanical view and get cut or clipped at the macro crop borders

are **NOT violations**. Slicing through a peripheral part bubble is a natural consequence of view boundary boxing and must be ignored during Pass 1.

- **A.7. Completeness and Zero Omissions.** You must operate with absolute exhaustiveness and completeness. Scan the entire page canvas systematically from top-left to bottom-right. Your goal is to ensure that **every single valid mechanical drawing view, project elevation, specifications table, BOM sheet, or technical notes block is successfully discovered and drafted**. Leaving valid structural units un-cropped is a critical failure. If a drawing view or table exists on the page, you must draft it.

V.A — Macro-Object Quality & Verification

These rules are designed specifically for the verifier and the re-crop agent, focusing on visual integrity, artifact completeness, and defect identification, structurally distinct from the discovery rules in Section A.

Your task is to rigorously inspect a proposed crop and determine if it represents a structurally complete, visually cohesive, and perfectly isolated machine-catalog artifact. You must discard crops that are damaged by slicing, truncation, or false selection, and guide the re-cropping process to fix them.

- **V.A.1. Structural Completeness & Geometry Truncation.** The cropped image must capture the *entirety* of the intended mechanical assembly, drawing view, or structural unit. You must **DISCARD** crops that slice through the main machine geometry, chop off extremities (like exhaust pipes, base frames, or electrical cabinets), or truncate major components belonging to the primary view. The structure must be visually cohesive and completely enclosed.
- **V.A.2. Table Boundary Integrity.** Bill of Materials (BOM) tables, specification sheets, and title blocks must be perfectly intact. You must **DISCARD** any crop that horizontally or vertically slices through table rows, text, or border lines. If tables or title blocks are stacked, they must either be captured together entirely without internal slicing, or cleanly separated.
- **V.A.3. Text Legibility & Note Isolation.** For standalone notes blocks or grouped text, every character and word must be fully legible. You must **DISCARD** crops that slice horizontally through words, cut off the top/bottom of text lines, or leave orphaned, illegible text fragments bleeding in at the boundaries.
- **V.A.4. Isolation and Bleeding.** The crop must cleanly isolate its target object from surrounding elements. You must **DISCARD** crops that suffer from severe "bleeding"—where large, truncated portions of an adjacent, unrelated

mechanical view or a separate table intrude significantly into the frame, causing visual confusion.

- **V.A.5. False Positives & Prose Rejection.** A valid crop must contain a cohesive technical structure (drawing, table, notes block). You must **PERMANENTLY DISCARD** (do not re-crop) any image that consists entirely of empty white space, decorative page borders, Table of Contents (TOC), or running paragraphs of prose narrative with no visual structures.
- **V.A.6. Permitted Peripheral Slicing Leniency.** Slicing through peripheral part balloon callouts (circled numbers) or stray leader lines that fall on the extreme outer borders of the crop is EXPECTED and ACCEPTABLE. You must **NOT DISCARD** a crop solely because a marginal balloon callout is cut in half, provided the main mechanical view itself is intact and fully captured.

B — Micro-Detail Subdivision (Pass 2)

Your task is to scan a pre-cropped, high-resolution Macro-Object drawing view (e.g., a front view committed in Pass 1) and subdivide it by extracting clean, granular annotation-geometry pairs.

- **B.1. Strict Target Limitation.** You must ONLY target complete, descriptive text boxes and multi-word text notes that explicitly have leader lines or arrows pointing to drawing components.
- **COMPLETELY IRRELEVANT AND STRICTLY FORBIDDEN FROM CROPPING:**
 - **Circled Balloon Callouts / Part Numbers:** Circle bubbles containing numbers pointing to parts are completely irrelevant and must be ignored. They are NOT valid targets.
 - **Callouts without Text:** Bare arrow lines or leader lines pointing to parts, lacking any multi-word descriptive text notes, are completely irrelevant and must be ignored.
 - **Geometrical Measurements / Dimensions:** Bare physical measurements and dimension arrows are completely irrelevant and must be ignored.
 - **Text or Cells Inside Tables:** Cropping individual text blocks, rows, or cells residing inside specifications tables, BOM tables, or parameter sheets is STRICTLY FORBIDDEN and must be ignored.
- **B.2. Trace Leader Line and Enclose Target Component.** Never crop the text note alone. A note floating alone on whitespace, with no referent machine geometry, is broken evidence.

- For each descriptive text note or box:
 - i. **Enumerate and Trace:** Trace every leader line exiting the text frame from its starting point to its arrowhead pointing to the machine geometry.
 - ii. **Capture the Triple:** Enlarge your coordinate box generously so that BOTH the text note (including its borders) AND the physical machine part or section of the drawing that its leader line points to are fully and cleanly inside the crop (the complete text, leader, target triple).
 - iii. **Provide Context:** Include enough of the surrounding machine drawing context so the reader can understand exactly what physical component the text note refers to. No tight text-only crops!
- **B.3. Refine centering.** Ensure no bias toward any corner, minimizing excess whitespace on the opposite side to keep focus sharp on the annotation-geometry pair.

V.B — Micro-Detail Quality & Verification

These rules govern the quality evaluation of granular annotation-geometry crops. Your task is to rigorously inspect a proposed crop to ensure the entire text note, its leader lines, and the full physical component it describes are completely intact. You must discard crops damaged by slicing or truncation, while explicitly permitting unavoidable background overlaps.

- **V.B.1. Text Frame Legibility & Completeness.** Every character, word, and the surrounding border of the primary descriptive text note must be completely intact. You must **DISCARD** crops that horizontally or vertically slice through the text box, cut off words, or clip the text frame boundaries.
- **V.B.2. Leader Line & Arrowhead Integrity.** The trace from the text note to the target component must be unbroken. You must **DISCARD** crops where the leader line exiting the text frame is clipped midway, or where the final arrowhead pointing to the machine geometry is truncated.
- **V.B.3. Target Geometry Completeness.** The physical machine part or structural section indicated by the arrowhead **MUST** be fully contained within the crop. You must **DISCARD** crops that slice through the target component, leaving only a fragment of the geometry visible. The reader must be able to see the complete physical context of the part being described.
- **V.B.4. Superposition & Density Leniency.** Because micro-details are extracted from dense, high-resolution drawing views, overlapping geometry is unavoidable. You must **NOT DISCARD** a crop simply because unrelated adjacent components, intersecting background lines, or peripheral part balloons bleed

into the frame. As long as the primary text, leader line, and target geometry triple are fully complete, background clutter and superposition are perfectly acceptable.

- **V.B.5. Centering & Unnecessary Whitespace.** The target annotation-geometry pair should be reasonably centered. You must **DISCARD** crops that capture massive, excessive areas of completely empty whitespace that push the target pair far into an extreme corner, severely reducing the available resolution of the actual subject.

C — Captioning + Vision Prompts

- **C.1. Scope — what you receive, what you produce, what to ignore.**
- **Input:** one rendered crop image. No other context.
- **Output:** two fields in this order — `vision_prompt` (paragraph), then `caption` (short label).
- **Anchor:** both fields describe the SAME main subject — the central label/subject the crop was made for.
- **Ignore peripheral text near the borders; partially-cut-off labels belong to other crops.**
- **C.2. Vision prompt — semantic meaning of the main subject.**
- **Form:** multi-sentence paragraph anchored on the crop's main annotation.
- **Steps:** identify the main text label (or text note); identify the object it indicates; explain what that object IS and what it does in the context of the larger drawing.
- **NOT a parts list** (e.g. "two valves, dimension lines, a hatched cylinder").
- **NOT a bare label** (e.g. "Heating element clearance").
- **Example** — label: "dismount space for electric heating element": > *"The crop highlights the label 'dismount space for electric heating element', which marks the unobstructed clearance required above an electric heating element that protrudes from the top of the cleaning machine. The element immersion-heats the washing tank water; its tubular shaft must periodically be lifted out vertically for inspection and de-scaling, which is why a clearance envelope of 1511.5 mm horizontally and 668 mm vertically must remain free above the machine (with 420 mm of lateral clearance). The dashed outline visualizes the element's swing path during removal."*
- **C.3. Caption — short semantic label for the same main subject.**

- **Form:** one sentence or a short name saying what the image IS.
- **Anchor:** refer only to the main subject identified in B.2; not everything visible.
- **Example:** "Electrical supply connection (3x400V 125A)".
- **Why it matters:** long descriptive captions are misleading and cause wrong layer classification.